

EMT Disconnected One-Point Building Blocks

Quark Loops, Gluon Loops, Stochastic Sources, and HDF5 Outputs

PyQUDA Measurement Notes

June 10, 2026

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1 Purpose and File Map

This note documents the hadron-independent one-point functions used to build disconnected contributions to proton and pion energy-momentum tensor (EMT) matrix elements. The main implementation is

`pyquda_measurement_utils/Disconnected_1pt_EMT_vibe_develop.py`.

The Perlmutter application entry points are in

`application/EMT_disconnected_1pt/perlmutter`.

The shorter command-oriented workflow guide is

`docs/EMT_disconnected_1pt/EMT_proton_disconnected_guide.md`.

The one-point workflows do not form hadron three-point functions by themselves. They measure vacuum loop building blocks. A disconnected hadron matrix element is formed later by correlating these one-point loops with hadron two-point functions on the same gauge ensemble.

2 Conventions

We work in Euclidean spacetime with four directions. Mathematical formulas use $\mu, \nu, \rho = 1, 2, 3, 4$, while Python loops use zero-based indices $\mu, \nu, \rho = 0, 1, 2, 3$. The temporal direction is the fourth Euclidean direction. PyQUDA stores local lattice fields in even-odd layout; the displayed formulas suppress this storage detail.

For a spatial momentum transfer q , the code constructs spatial phases with `MomentumPhase.getPhases`. In this note the phase is written as

$$\Phi_q(x) = e^{i\mathbf{q}\cdot\mathbf{x}}, \quad (1)$$

where only the spatial coordinates are Fourier transformed and the time coordinate t is kept explicit.

The spatial volume normalization used in the averaged HDF5 datasets is

$$V_3 = L_x L_y L_z. \quad (2)$$

The raw per-source quark datasets are stored before this V_3 normalization. The averaged quark and gluon datasets are volume-normalized.

3 Quark One-Point Workflow

The quark workflow is usually the first part to study. It contains the stochastic trace estimator, the hierarchical-probing source bookkeeping, the χ -type diagnostics, and the flowed quark EMT loop used in disconnected analysis.

3.1 Target Trace

A disconnected quark loop has the schematic form

$$L_\Gamma^q(q, t; t_f) = \sum_{\mathbf{x}} \Phi_q(x) \text{Tr}_{sc}[\Gamma[U(t_f)](x) D^{-1}[U](x, x)], \quad (3)$$

where D^{-1} is the quark propagator, Γ is the local operator kernel, and Tr_{sc} denotes spin-color trace. Directly computing all diagonal entries of $D^{-1}(x, x)$ is too expensive, so the trace is estimated with stochastic sources.

3.2 Stochastic Trace Estimator

Let ξ_r be a stochastic source vector whose spin-color-site components satisfy

$$\langle \xi_r \rangle_{\text{noise}} = 0, \quad \left\langle \xi_r(i) \xi_r^\dagger(j) \right\rangle_{\text{noise}} = \delta_{ij}. \quad (4)$$

Here i, j are combined site, spin, and color indices. The code obtains

$$\eta_r = D^{-1} \xi_r \quad (5)$$

from the Dirac inverter. For any operator matrix A ,

$$\left\langle \xi_r^\dagger A \eta_r \right\rangle_{\text{noise}} = \left\langle \xi_r^\dagger A D^{-1} \xi_r \right\rangle_{\text{noise}} \quad (6)$$

$$= \sum_{i,j,k} A_{ij} (D^{-1})_{jk} \left\langle \xi_{r,i}^\dagger \xi_{r,k} \right\rangle_{\text{noise}} \quad (7)$$

$$= \sum_{i,j} A_{ij} (D^{-1})_{ji} = \text{Tr}(A D^{-1}). \quad (8)$$

This is the Hutchinson trace-estimator logic used by the quark one-point workflow.

3.3 Z_n Noise

The ordinary stochastic scheme is selected with

$$\text{noise_scheme} = \text{zn}. \quad (9)$$

Each component of ξ is drawn independently from

$$\xi(i) \in \left\{ \exp\left(\frac{2\pi i r}{n}\right) \mid r = 0, 1, \dots, n-1 \right\}, \quad (10)$$

where $n = \text{n.zn}$. The default smoke-test choice is usually $\mathbb{Z}_2$ noise.

3.4 Quark EMT Operator Implemented in Code

The quark EMT kernel in the one-point code is the local bilinear with a covariant derivative acting on the solution vector η . The implemented per-source, pre-volume-normalized object is

$$L_{\nu\mu}^{q,(r)}(q, t; t_f) = -\frac{1}{2} \sum_{\mathbf{x}} \Phi_q(x) \xi_r^\dagger(x; t_f) \gamma_\nu \left[D_{+\mu} - D_{-\mu} \right] \eta_r(x; t_f). \quad (11)$$

This formula maps directly to `_get_Tmunu_symmetrized_P_Breit_slice`. In code:

```
tmp = U_f.pure_gauge.covDev(eta, mu) - U_f.pure_gauge.covDev(eta, mu + 4)
Y = contract("ab,...bc->...ac", D_gammas_local[nu], tmp.data)
complex_field = contract("...sc,...sc->...", xi.data.conj(), Y)
Tmunu[nu, mu] += -0.5 * impose_P_Breit_slice(...)
```

The first tensor index is ν , because the gamma matrix γ_ν is applied after choosing the derivative direction μ . This is why the code fills `Tmunu[nu, mu]` before symmetrization.

3.5 Symmetrization

The EMT tensor is symmetrized after all unsymmetrized components have been accumulated:

$$L_{\mu\nu}^q \leftarrow \frac{1}{2} (L_{\mu\nu}^q + L_{\nu\mu}^q). \quad (12)$$

The code then sets both entries equal. The HDF5 writer stores only the upper triangle $\mu \leq \nu$, with dataset names T11, T12, ..., T44. The lower triangle should be reconstructed by symmetry if needed.

3.6 Noise Average and Volume Normalization

For N_{eff} effective sources, the averaged quark one-point loop is

$$L_{\mu\nu,\text{avg}}^q(q, t; t_f) = \frac{1}{V_3} \frac{1}{N_{\text{eff}}} \sum_{r=1}^{N_{\text{eff}}} L_{\mu\nu}^{q,(r)}(q, t; t_f). \quad (13)$$

The code stores both the raw per-source data and the averaged volume-normalized data:

$$\text{raw/Tmunu_pervec} \leftrightarrow L_{\mu\nu}^{q,(r)}(q, t; t_f), \quad (14)$$

$$\text{avg/Tmunu/Tij} \leftrightarrow L_{ij,\text{avg}}^q(q, t; t_f). \quad (15)$$

3.7 χ Diagnostics

The quark workflow also writes two scalar diagnostics:

$$\chi_0^{(r)}(q, t; t_f) = \sum_{\mathbf{x}} \Phi_q(x) \xi_r^\dagger(x; t_f) \eta_r(x; t_f), \quad (16)$$

$$\chi_1^{(r)}(q, t; t_f) = \sum_{\mathbf{x}} \Phi_q(x) \xi_r^\dagger(x; t_f) \xi_r(x; t_f). \quad (17)$$

χ_0 is the stochastic scalar trace estimate. χ_1 checks the noise normalization. In code these are the two entries of CHI:

```
dot_xi_eta = contract("etzyxbc,etzyxbc->etzyx", xi.data.conj(), eta.data)
CHI[0] = impose_P_Breit_slice(U_f, dot_xi_eta, phases_3pt)
```

```
dot_xi_xi = contract("etzyxbc,etzyxbc->etzyx", xi.data.conj(), xi.data)
CHI[1] = impose_P_Breit_slice(U_f, dot_xi_xi, phases_3pt)
```

The averaged HDF5 dataset avg/CHI is divided by V_3 , just like the averaged quark EMT datasets.

3.8 Ringed-Fermion Kinetic Normalization

The raw CHI datasets are diagnostics, not the ringed-fermion normalization itself. The flowed ringed-fermion normalization uses the flowed fermion kinetic expectation value. In this implementation, the relevant building block is reconstructed from the zero-momentum diagonal quark EMT components:

$$\sum_{\mu=1}^4 L_{\mu\mu,\text{avg}}^q(q=0, t; t_f) = -\frac{1}{2V_3} \sum_{\mu, \mathbf{x}} \xi^\dagger(x; t_f) \gamma_\mu [D_{+\mu} - D_{-\mu}] \eta(x; t_f). \quad (18)$$

In HDF5 this is obtained from avg/Tmunu/T11, T22, T33, and T44 at the $q=0$ momentum index.

3.9 Gradient-Flow Schedule for Quarks

The quark workflow measures before flowing. Therefore step 0 is the unflowed observable. If $\epsilon = \text{flow_epsilon}$, then

$$t_f(n) = n \epsilon, \quad n = 0, \dots, \text{flow_steps}. \quad (19)$$

The first flow interval is internally split into ten smaller updates:

```

for step in range(Nsteps + 1):
    measure Tmunu and CHI
    if step == 0:
        flow [xi, eta] for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow [xi, eta] for 1 step of stepsize

```

The stochastic source ξ and solution η are flowed together as one `MultiField`. This keeps both fields on the same flowed gauge background.

4 Hierarchical Probing and Source Bookkeeping

The quark one-point workflow supports

$$\text{noise_scheme} \in \{\text{zn}, \text{hierarchical_probing}\}. \quad (20)$$

Hierarchical probing starts from a base noise vector ξ_b and multiplies it by deterministic site-only Rademacher vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x) \xi_b(x). \quad (21)$$

The probing vectors are designed to reduce near-diagonal contributions to the stochastic trace variance. In the current implementation the number of HP vectors must be a positive power of two:

$$\text{hp_num_vectors} = 1, 2, 4, 8, \dots \quad (22)$$

The effective number of inversions is

$$N_{\text{eff}} = \begin{cases} \text{n_vec}, & \text{noise_scheme}=\text{zn}, \\ \text{n_vec} \times \text{hp_num_vectors}, & \text{noise_scheme}=\text{hierarchical_probing}. \end{cases} \quad (23)$$

This value is stored as `effective_n_inversions`.

4.1 Ordering Choices

Two HP orderings are currently implemented:

Ordering	Meaning
<code>global_xyz_t_gray_projected_to_evenodd</code>	Gray-code the full x, y, z, t site id.
<code>spatial_xyz_then_t_gray_projected_to_evenodd</code>	Gray-code spatial sites first, then append time bits.

The spatial-first option is useful when early HP vectors should emphasize spatial locality before time locality.

4.2 Bookkeeping Datasets

For each effective source, the output records:

Dataset	Meaning
<code>raw/source_index</code>	Effective source index after HP expansion.
<code>raw/base_noise_index</code>	Original base noise index.
<code>raw/hp_index</code>	HP vector index used for that base noise.

For example, with `n_vec = 1` and `hp_num_vectors = 2`, the expected bookkeeping is

$$\text{source_index} = [0, 1], \quad \text{base_noise_index} = [0, 0], \quad \text{hp_index} = [0, 1]. \quad (24)$$

5 Quark HDF5 Layout

The quark writer is `save_empt_quark_1pt_hdf5`. The main raw datasets are

$$\text{raw/Tmunu_pervec} : [N_{\text{eff}}, 4, 4, N_q, N_{\text{steps}} + 1, N_t], \quad (25)$$

$$\text{raw/CHI_pervec} : [N_{\text{eff}}, 2, N_q, N_{\text{steps}} + 1, N_t]. \quad (26)$$

The averaged datasets are

$$\text{avg/Tmunu/T11, T12, } \dots, \text{T44} : [N_q, N_{\text{steps}} + 1, N_t], \quad (27)$$

$$\text{avg/CHI} : [2, N_q, N_{\text{steps}} + 1, N_t]. \quad (28)$$

Only $\mu \leq \nu$ tensor components are written in the averaged `avg/Tmunu` group.

The exact quark dataset and attribute names used by downstream scripts are:

```
raw/Tmunu_pervec
raw/CHI_pervec
raw/source_index
raw/base_noise_index
raw/hp_index
avg/Tmunu/T11
avg/Tmunu/T12
avg/Tmunu/T22
avg/Tmunu/T33
avg/Tmunu/T44
avg/CHI
noise_scheme
hp_num_vectors
effective_n_inversions
```

Important attributes include

Attribute	Meaning
<code>measurement</code>	<code>quark_1pt</code> .
<code>flow_type, flow_epsilon, flow_steps, flow_times</code>	Gradient-flow schedule.
<code>qext, pf, p_2pt</code>	Momentum labels used by downstream analysis.
<code>volume_norm</code>	V_3 , the spatial-volume normalization.
<code>upper_triangle_only</code>	Whether only $\mu \leq \nu$ components are written.
<code>n_vec, n_base_noise, effective_n_inversions</code>	Stochastic source counts.
<code>noise_scheme, hp_num_vectors, hp_ordering</code>	Source-generation scheme.

6 Gluon One-Point Workflow

The gluon workflow is independent of stochastic quark sources. It constructs a flowed gauge-field strength and then measures a spatially Fourier-projected gluon EMT building block.

6.1 Clover Field Strength

For each independent plane (μ, ν) , the code constructs the four oriented clover plaquettes

$$\mu \rightarrow \nu \rightarrow -\mu \rightarrow -\nu, \quad (29)$$

$$\nu \rightarrow -\mu \rightarrow -\nu \rightarrow \mu, \quad (30)$$

$$-\mu \rightarrow -\nu \rightarrow \mu \rightarrow \nu, \quad (31)$$

$$-\nu \rightarrow \mu \rightarrow \nu \rightarrow -\mu. \quad (32)$$

Let $Q_{\mu\nu}(x)$ be their sum. The anti-Hermitian clover field is

$$A_{\mu\nu}(x) = \frac{1}{8} \left[Q_{\mu\nu}(x) - Q_{\mu\nu}^\dagger(x) \right]. \quad (33)$$

The code projects this matrix to the traceless $\mathfrak{su}(3)$ algebra:

$$A_{\mu\nu}^{\text{tl}}(x) = A_{\mu\nu}(x) - \frac{1}{N_c} \text{Tr}_c[A_{\mu\nu}(x)] \mathbf{1}. \quad (34)$$

Finally, the Euclidean field-strength convention used by the implementation is

$$F_{\mu\nu}(x) = -i A_{\mu\nu}^{\text{tl}}(x). \quad (35)$$

Only the six independent planes are computed, and the antisymmetric entries are filled with

$$F_{\nu\mu}(x) = -F_{\mu\nu}(x). \quad (36)$$

The traceless projection is a field-strength projection. It is not a traceless projection of the final EMT tensor.

6.2 Implemented Gluon EMT Loop

The implemented gluon one-point building block is

$$L_{\mu\nu}^g(q, t; t_f) = \frac{2}{V_3} \sum_{\mathbf{x}} \Phi_q(x) \sum_{\substack{\rho=1 \\ \rho \neq \mu, \nu}}^4 \text{Tr}_c[F_{\mu\rho}(x; t_f) F_{\nu\rho}(x; t_f)]. \quad (37)$$

This maps directly to the code in `EMTDisconnectedGluon1pt.flowed_1pt`:

```
for mu in range(4):
    for nu in range(mu, 4):
        tmp = zeros(...)
        for rho in range(4):
            if rho == mu or rho == nu:
                continue
            tmp += contract("...ab,...ba->...", F[mu][rho], F[nu][rho])
        Tmunu_t[mu, nu, :, step, :] += 2.0 * slice_t
Tmunu_t /= Ns3
```

The factor 2 is explicit in the code. The division by V_3 is applied after all spatial Fourier sums have been accumulated.

6.3 Gradient-Flow Schedule for Gluons

As in the quark workflow, step 0 is measured before flow and therefore is the unflowed gauge observable. After each measurement, the gauge field is advanced by Wilson or Symanzik flow:

```
for step in range(Nsteps + 1):
    measure F and Tmunu
    if step == 0:
        flow gauge for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow gauge for 1 step of stepsize
```

7 Gluon HDF5 Layout

The gluon writer is `save_empt_gluon_1pt_hdf5`. Unlike the quark file, there are no stochastic raw per-vector datasets. The output contains only volume-normalized upper-triangle tensor components:

$$\text{Tmunu}/\text{T11}, \text{T12}, \dots, \text{T44} : [N_q, N_{\text{steps}} + 1, N_t]. \quad (38)$$

Important attributes include

Attribute	Meaning
<code>measurement</code>	<code>gluon_1pt</code> .
<code>flow_type</code> , <code>flow_epsilon</code> , <code>flow_steps</code> , <code>flow_times</code>	Gradient-flow schedule.
<code>qext</code> , <code>pf</code> , <code>p_2pt</code>	Momentum labels used by downstream analysis.
<code>volume_norm</code>	V_3 , the spatial-volume normalization.
<code>upper_triangle_only</code>	Whether only $\mu \leq \nu$ components are written.

8 Disconnected Combination with Proton Two-Point Functions

The quark and gluon one-point loops are hadron independent. They become a proton disconnected three-point building block only after being correlated with the proton two-point function on the gauge ensemble. For a proton two-point function $C_2^p(p_f, t)$, define

$$C_{3,\mu\nu}^{p,\text{disc}}(p_f, p_i; t, \tau; t_f) = \langle C_2^p(p_f, t) L_{\mu\nu}(q, \tau; t_f) \rangle_{\text{cfg}} - \langle C_2^p(p_f, t) \rangle_{\text{cfg}} \langle L_{\mu\nu}(q, \tau; t_f) \rangle_{\text{cfg}}, \quad (39)$$

with

$$q = p_f - p_i. \quad (40)$$

The subtraction term removes the vacuum expectation value of the one-point loop. This subtraction must be done at the ensemble-analysis level using the same configuration set as the proton two-point function.

A basic ratio is

$$R_{\mu\nu}^{p,\text{disc}}(t, \tau; t_f) = \frac{C_{3,\mu\nu}^{p,\text{disc}}(p_f, p_i; t, \tau; t_f)}{C_2^p(p_f, t)}. \quad (41)$$

Kinematic projectors, EMT mixing coefficients, and gradient-flow matching factors are applied later in the physics analysis.

9 End-to-End Proton Diagnostic Workflow

The application directory

`application/EMT_disconnected_1pt/perlmutter`

contains a minimal four-step diagnostic workflow:

1. measure the stochastic quark one-point loop;
2. measure the gluon one-point loop;
3. measure the proton two-point function C_2^p ;
4. combine C_2^p and the one-point loops into disconnected diagnostic three-point objects.

For the quark loop, the builder reads `raw/Tmunu_pervec` and forms cumulative stochastic averages

$$\bar{L}_{\mu\nu}^{q,N}(q, t; t_f) = \frac{1}{V_3} \frac{1}{N} \sum_{r=1}^N L_{\mu\nu}^{q,(r)}(q, t; t_f), \quad N = 1, \dots, N_{\text{eff}}. \quad (42)$$

This is the quantity used to check whether the stochastic estimator changes smoothly as more random sources are included.

For a single configuration, the builder can only form the unsubtracted proxy

$$C_{3,\mu\nu}^{\text{unsub},N}(t_{\text{sep}}, q, t; t_f) = C_2^p(t_{\text{sep}}) \bar{L}_{\mu\nu}^{q,N}(q, t; t_f). \quad (43)$$

This object is useful for checking shapes, source-count dependence, time alignment, and HDF5 plumbing. It is not the physical disconnected correlator, because the vacuum subtraction in Eq. (39) requires an ensemble average.

With multiple configurations, the builder applies the ensemble subtraction for each cumulative source count:

$$C_{3,\mu\nu}^{\text{disc},N} = \left\langle C_2^p \bar{L}_{\mu\nu}^{q,N} \right\rangle_{\text{cfg}} - \langle C_2^p \rangle_{\text{cfg}} \left\langle \bar{L}_{\mu\nu}^{q,N} \right\rangle_{\text{cfg}}, \quad (44)$$

$$R_{\mu\nu}^{\text{disc},N} = \frac{C_{3,\mu\nu}^{\text{disc},N}}{\langle C_2^p \rangle_{\text{cfg}}}. \quad (45)$$

The gluon loop has no stochastic source axis, so the builder forms the same unsubtracted and vacuum-subtracted objects without a cumulative N index.

The diagnostic merger output contains:

```
C2
quark/source_count
quark/loop_cumulative
quark/C3_unsubtracted_cumulative
quark/C3_disc_cumulative      # only if Ncfg >= 2
quark/R_disc_cumulative      # only if Ncfg >= 2
gluon/loop
gluon/C3_unsubtracted
gluon/C3_disc                 # only if Ncfg >= 2
gluon/R_disc                 # only if Ncfg >= 2
summary/quark_source_count
summary/quark_T44_q0_flow0_loop_norm
summary/quark_T44_q0_flow0_unsub_ratio_proxy
```

10 How to Read Smoke Tests

A single S8T32 test gauge checks implementation conventions, not physics signals. Useful smoke-test checks are:

- the code runs on the intended backend;
- `flow_times` contains step 0 and the requested flow times;
- `upper_triangle_only=True`;
- quark raw datasets have the expected $[N_{\text{eff}}, 4, 4, N_q, N_{\text{steps}} + 1, N_t]$ and $[N_{\text{eff}}, 2, N_q, N_{\text{steps}} + 1, N_t]$ shapes;
- HP runs have consistent `raw/source_index`, `raw/base_noise_index`, and `raw/hp_index`;
- gluon datasets contain `Tmunu/T11` through `T44` but no stochastic `raw/Tmunu_pervec`.

11 Recommended Next Algorithmic Upgrade

The current variance-reduction options are ordinary Z_n noise and hierarchical probing. A planned future direction is multiplier-based graph coloring, motivated by

arXiv:2606.00394.

The motivation is that standard HP naturally tests 2^k source counts, whereas multiplier-based coloring can test intermediate coloring budgets. A future scheme should be added as a separate optional source scheme, with HDF5 metadata that clearly distinguishes it from `hierarchical_probing`.